ARIMA, VAR & Multivariate Models

ECON 8310: Business Forecasting — Lecture 2

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Lecture Outline

Stationarity and Differencing

Before fitting any model, the series must have a stable mean and variance.

What Is Stationarity?

Weak (Covariance) Stationarity

A time series $\{y_t\}$ is **weakly stationary** if:

1. $\mathbb{E}[y_t] = \mu$ (constant mean over time)
2. $\mathrm{Var}(y_t) = \sigma^2 < \infty$ (constant, bounded variance)
3. $\mathrm{Cov}(y_t, y_{t-k}) = \gamma_k$ depends only on lag k, not on t

Why the definition earns its keep: if the mean drifts, there is no single value to forecast. If the variance grows, prediction intervals widen without bound. Stationarity is what makes the past informative about the future — it is the assumption every model in this lecture rests on.

The usual culprits are trending series — GDP, prices, sales — and variance that grows with the level. First-differencing removes a stochastic trend; a log transform stabilizes variance.

Rule of thumb: visible trend or growing variance $\Rightarrow$ likely non-stationary. Test formally with ADF or KPSS.

Unit Roots and Differencing

AR(1): $y_t = \phi_1 y_{t-1} + \varepsilon_t$. Everything turns on whether ϕ_1 is below one or exactly one.

Condition	Behavior	Implication
$\lvert\phi_1\rvert < 1$	Shock decays, reverts to mean	Stationary AR(1)
$\phi_1 = 1$	Shock persists *permanently*	Random walk (unit root)

First differencing removes a stochastic trend: $\Delta y_t = y_t - y_{t-1}$. So $d = 0$ means already stationary, $d = 1$ means one first difference, and $D = 1$ means a seasonal difference, $\Delta_{12} y_t = y_t - y_{t-12}$.

Testing: ADF (H_0: unit root) and **KPSS** (H_0: stationary) test *opposite* nulls — so run both, and be suspicious when they disagree. **Do not over-difference.** A second difference on an already-stationary series induces negative autocorrelation that you then have to model away — you have manufactured the problem you are now solving. Test *after* differencing, and stop at $d = 1$ once the result is stationary.

ACF and PACF: Model Identification

The **autocorrelation function**, $\rho_k = \mathrm{Corr}(y_t, y_{t-k})$, measures how y_t relates to its own past. The **partial** autocorrelation measures the same thing while controlling for the lags in between — which is what separates a direct effect from one inherited through intermediate lags.

Read them together and they identify the model:

Model	**ACF**	**PACF**
AR(p)	Tails off geometrically	Cuts off after lag p
MA(q)	Cuts off after lag q	Tails off geometrically
ARMA	Tails off	Tails off
RW / I(1)	Decays slowly	Big spike at lag 1

Always unit-root test and difference *before* reading either plot. The ACF of a non-stationary series decays slowly no matter what the underlying process is, so it tells you nothing about p or q — only that you have not differenced yet.

ARIMA Models

Combine differencing with AR and MA components into one unified model.

ARIMA(p,d,q)

> **ARIMA(p,d,q) Model**
>
> Apply an ARMA(p,q) to the d-times differenced series: $y_t = c + \sum_{i=1}^{p} \phi_i y_{t-i} + \varepsilon_t + \sum_{j=1}^{q} \theta_j \varepsilon_{t-j}$

p is the number of AR lags, d the number of differences, q the number of MA lags. Several familiar models are just corners of this space: **ARIMA(0,1,0)** is a random walk, **ARIMA(1,0,0)** is an AR(1), and **ARIMA(0,1,1)** is algebraically equivalent to simple exponential smoothing from Lecture 1.

A forecast by hand. Take ARIMA(1,1,0) with $\hat{\phi}_1 = 0.5$. If $y_{100} = 120$ and $y_{99} = 118$, then $\Delta y_{100} = 2$, and

$$\hat{y}_{101|100} \;=\; 120 + 0.5 \times 2 \;=\; \mathbf{121}.$$

In Python: `pmdarima.auto_arima()` searches orders automatically; `statsmodels.SARIMAX` fits any order you specify.

SARIMA: Adding Seasonal Terms

For monthly or quarterly data, autocorrelation often appears at lags $m, 2m, 3m, \ldots$ SARIMA(p,d,q)(P,D,Q)[m] adds seasonal AR and MA polynomials alongside the non-seasonal ones.

The useful mental model is **two ARIMA models running at different time scales**: the non-seasonal part (p, d, q) captures week-to-week dynamics, while the seasonal part $(P, D, Q)[m]$ captures the year-over-year pattern at lag m.

Practical workflow:

1. ADF / KPSS $\rightarrow$ choose d
2. Seasonal unit root test $\rightarrow$ choose D
3. ACF / PACF on the differenced series $\rightarrow$ guide p, q, P, Q
4. Minimize AIC, then check the residual ACF with a Ljung-Box test

For monthly retail data with trend and December seasonality, SARIMA(1,1,1)(0,1,1)[12] is a strong starting point — and a good default to beat before trying anything more elaborate.

VAR Models

When two series influence each other, model them together symmetrically.

Why Use Multiple Series?

Every model so far — ETS, ARIMA — uses only y_t's own history. But business variables are linked: consumer sentiment today moves retail spending next month, interest rates move housing starts within two quarters, advertising spend moves sales over the following weeks.

The principle underneath all of it: **if x_t carries information about y_{t+h} that is not already in y_t's own history, including x_t reduces forecast error. Granger causality** (Granger 1969) is the formal test of exactly that claim, and we get to it in two slides.

That leaves a modelling choice, and it turns on whether the influence runs one way or both:

VAR — symmetric

- Every variable predicts every other
- Use when influence runs both ways
- No variable is privileged

ARIMAX — one direction

- x_t drives y_t, not the reverse
- Use when the causal story is clear
- Needs a forecast of x_t itself

The VAR(p) Model

For two stationary series y_{1t} and y_{2t}, each equation regresses on the lags of *both*:

$$\begin{cases} y_{1t} = c_1 + a_{11}\, y_{1,t-1} + a_{12}\, y_{2,t-1} + \varepsilon_{1t} \\ y_{2t} = c_2 + a_{21}\, y_{1,t-1} + a_{22}\, y_{2,t-1} + \varepsilon_{2t} \end{cases}$$

The cross terms are where the value is: a_{12} is how much $y_{2,t-1}$ predicts y_{1t}, which is precisely what a univariate AR throws away. If $a_{12} = a_{21} = 0$, the system collapses back into two separate AR models and you have learned that too. Each equation is estimated by OLS; the order p is chosen by BIC.

> **Parameter explosion.** A VAR(p) with k variables has $k^2 p$ coefficients. At $k = 5$ and $p = 4$ that is **100 coefficients** — typically more than a monthly business series has observations. Keep $k \leq 4$ unless you are using a LASSO-VAR.

Python: `statsmodels.tsa.api.VAR`; `var.fit(maxlags=p, ic='bic')` selects the order.

Granger Causality Test

Granger Causality

x_t **Granger-causes** y_t if past values of x_t help predict y_t *beyond what* y_t*'s own past already explains*.

The test: fit the y-equation with and without the x-lags, then run an F-test of H_0: all x-lag coefficients are zero. Rejecting means x_t Granger-causes y_t.

A concrete question. Does consumer sentiment (`UMCSENT`) Granger-cause retail sales (`RSXFS`)? The null is that every sentiment lag is zero in the retail equation. We will run exactly this in the lab.

Granger causality is not structural causality. It tests *predictive content*, not whether x mechanically brings y about — a rooster reliably predicts sunrise. Report it as "predictive of," never "causes."

Impulse response functions trace how a shock propagates through the system over time.

ARIMAX: Exogenous Variables with ARIMA Errors

When one variable clearly *drives* another rather than the relationship being symmetric, use ARIMAX.

ARIMAX Model

$$y_t = \mathbf{x}_t'\boldsymbol{\beta} + \eta_t, \qquad \eta_t \sim \mathrm{ARIMA}(p, d, q)$$

$\mathbf{x}_t$ are exogenous regressors; η_t absorbs whatever autocorrelation is left in the residuals.

Use it when the causal direction is clear (advertising $\rightarrow$ sales), you already have x_t or can forecast it, and adding x_t actually reduces cross-validated forecast error — check that last one rather than assuming it.

The catch that surprises people: to forecast y_{t+h} you need x_{t+h} — a forecast of your own predictor. If that is unavailable, ARIMAX buys you nothing at forecast time and you should use a VAR instead, which generates its own.

Python: `SARIMAX(y, exog=X, order=(p,d,q)).fit()`

Key Takeaways and Roadmap

Stationarity, ARIMA, VAR, and ARIMAX form the classical time series toolkit.

Key Takeaways

1. **Stationarity** is required before fitting time series models. Test with ADF (H_0: unit root) and KPSS (H_0: stationary). First-differencing removes a stochastic trend.
2. **ACF** identifies MA order; **PACF** identifies AR order. Always inspect after differencing, not before.
3. **ARIMA(p,d,q)** combines differencing with AR/MA. SARIMA adds seasonal polynomials for periodic autocorrelation. Use `auto_arima` for automatic selection.
4. **VAR** models all variables symmetrically and captures cross-variable dynamics. Keep $k \leq 4$. Use **Granger causality** to test predictive content.
5. **ARIMAX** adds exogenous predictors to ARIMA. Use when causal direction is clear and x_t is forecastable.

Next: Generalized Additive Models (GAMs) — flexible, interpretable nonlinear forecasting with Prophet and pyGAM.

References I

Granger, C. W. J. (1969). "Investigating Causal Relations by Econometric Models and Cross-Spectral Methods". In: *Econometrica* 37.3, pp. 424–438.